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Crack detection and characterization techniques—An overview

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SUMMARY

Crack occurrence and propagation are among critical factors that affect the performance and lifespan of civil infrastructures such as bridges, pipelines, and so on. As a consequence, numerous crack detection and characterization techniques have been researched and developed in the past decades in the areas of SHM and non-destructive evaluation (NDE). The significant amount of performed studies and the large number of publications give rise to the need to systematize, condensate, and summarize this enormous effort. The aims of this paper are to summarize the knowledge about cracking and its sources, review both existing and emerging methods for crack detection and characterization, and identify the advantages and challenges for these methods. In general, this paper identifies two sensing approaches (direct and indirect) and two data analysis approaches (model-based and model-free or data-driven) along with a range of associated technologies. The advantages and challenges of each approach and technology are discussed and summarized, and the future research needs are identified. This paper is intended to serve as a reference for researchers who are interested in crack detection and characterization as well as for those who are generally interested in SHM and NDE. Copyright © 2014 John Wiley & Sons, Ltd.

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KEY WORDS: crack detection and characterization; crack typology; structural health monitoring; non-destructive evaluation; direct sensing; model-based and model-free data analysis

1. INTRODUCTION

Civil structures and infrastructures provide essential welfare in society. Therefore, they are invaluable assets that need to be preserved and well maintained. In order to achieve this goal, it is necessary to assess their health conditions. A particular challenge is early-crack detection and characterization (localization and quantification), because the size of the crack is small at that stage. Cracks can be defined as unintentional discontinuities in structural materials. In general, they are the consequence of local material failure. They can be generated by many factors, such as external loads, buckling, fatigue, physical processes, such as freeze-thaw cycles, and chemical processes, such as alkali–silica reaction and corrosion. Typically, the cracks that require attention usually initiate with the opening size of 0.05–0.1 mm (0.002–0.004 in) approximately; they become of true concern at approximately 0.3–0.4 mm (0.013–0.016 in) and can develop to dangerous sizes of several millimeters to several tens of millimeters. While small cracks may lead to inadequate serviceability, large cracks can result in structural failures. Early-age cracking in concrete can seriously affect a structure's durability. In brittle materials, the occurrence of cracks may lead to immediate failure. However, in ductile materials, crack growth is usually gradual. It is important to detect all types of cracks before they reach a critical level and cause the failure of structural components or entire structures. Currently applied methods for crack detection and characterization include periodic visual inspections and

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non-destructive evaluation (NDE), which is based on traditional instrumentation and approaches [1]. Traditional methods can be expensive, time consuming, and frequently inaccurate during early stages of cracking by failing to detect damage. Visual inspections can be unreliable because some types of cracks are sparse or inconspicuous. Furthermore, if a crack occurs between two inspections, it may propagate and reach a critical size before it can be properly detected and characterized. Therefore, there is a need for more accurate, reliable, and affordable methods for quasi-real-time crack detection and characterization. Quasi-real-time assessment can be achieved through continuous monitoring.

A crack not only changes the local strain field in the structure but also affects other properties such as electrical conductivity, wave propagation properties, and so on. Cracks can change the global strain-field distribution, natural frequency, and the shape of the structure. Additionally, the initiation and propagation of cracks may generate acoustic waves. All these parameters can be, to a certain extent, used to detect and characterize the cracks. The most frequently applied techniques are based on strain sensors (electrical or fiber-optic), piezoelectric and acoustic sensors (acoustic emission and wave propagation monitoring), and accelerometers (modal analysis). Emerging approaches include micro-electromechanical systems (MEMS), piezoelectric paints, nano-technologies, large-area electronics (LAE), and laser-based noncontact sensing techniques. In addition, several data analysis algorithms can be combined with sensing technologies in order to detect and characterize cracks.

This paper is intended to serve as a reference for researchers who are interested in crack detection and characterization and/or general SHM and NDE methods. The main focus is on civil structures and infrastructure. It first presents a typology of cracks, including their causes, manifestations, and consequences. The following sections contain summaries of currently available sensing technologies and data analysis algorithms, identifying the advantages and challenges of each method. The paper ends with an overview of emerging technologies and conclusions.

2. CRACK CLASSIFICATION, CAUSES, AND CONSEQUENCES

2.1. Crack classification

Concrete (reinforced and prestressed) and steel are the most widely used construction materials in practice. Because both materials can experience cracking due to various causes, this paper focuses primarily on cracks in concrete and steel. A general method of classifying cracks is to identify them on the basis of their geometry [2]. On the basis of its direction of propagation, appearance, and position, a given crack in a beam structure can be classified as follows [2]:

- *transverse cracks* are perpendicular to the longitudinal axis of the beam structure;
- *longitudinal cracks* are parallel to the longitudinal axis of the beam structure;
- *slant cracks* have an angle with respect to the longitudinal axis of the beam structure;
- *breathing cracks* open when subjected to tensile stress and close when the stress is reversed;
- *gaping cracks*, also known as *notches*, remain open and do not close;
- *surface cracks* occur on the surface of the structure and can commonly be detected by visual inspection (except for buried, coated, or remote structures) or surface-mounted sensors; and
- *Subsurface cracks* propagate below the surface of the structure and can be detected by acoustic emission or wave propagation monitoring methods.

Transverse cracks are the most common and most dangerous cracks, because they can reduce a structure's cross section and therefore lower its structural capacity. Although longitudinal and slant cracks are less common, they can affect a structure's torsional capacity. Transverse breathing cracks may affect structures in the same manner as regular transverse cracks; that is the reason why many researchers focus on this type of crack. On the other hand, transverse gaping cracks are the easiest to simulate under laboratory conditions; therefore, most experimental studies involve gaping cracks. Research has indicated that with regard to the dynamic behavior of structures, surface cracks are more important than subsurface cracks [3]. Surface cracks often facilitate adverse chemical leaching and can accelerate material deterioration, especially in concrete structures.

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2.2. Cracks caused by mechanical actions

In general, cracks can be caused by mechanical actions (e.g., static, dynamic, and cyclic loadings) or by physical and chemical processes (e.g., shrinkage, thermal effects, corrosion, alkali–silica reaction, etc.).

Mechanically generated cracks can occur from several causes and vary in different materials. For concrete and steel, typical mechanisms include the following:

- cracking due to excessive external static or dynamic loadings (e.g., the material reaches critical levels of stress or strain);
- cracking due to repetitive or cyclic dynamic loading (e.g., fatigue);
- cracking due to stress concentrations (e.g., at the location of abrupt cross-section changes, dents, grooves, inclusions, forging flaws, material porosities, voids, etc.); and
- cracking due to buckling (e.g., [4]).

A typical sequence of progressive mechanical cracks leading to structural failure is described as follows [2]:

Crack initiation stage. In this stage, minute discontinuities occur in the material. Cracks can initiate from any or a combination of the aforementioned mechanisms. During the initiation stage, crack openings typically remain below 100 μm (0.004 in) in width and are thus, in the most cases, invisible to the human eye.

Crack propagation stage. In this stage, the initial discontinuities grow in size and propagate along a certain orientation. This is mainly caused by the variation of external loadings and environmental factors such as increased stresses, fatigue due to cyclic stresses, and changes in temperature and moisture. With respect to the structural capacity of reinforced concrete, crack widths of concern are usually about 0.33 mm (0.013 in) for internal exposure and 0.41 mm (0.016 in) for external exposure [5]. Nuclear reactors or waste treatment plants, however, have stricter regulations on crack size control. Conditions that can accelerate the crack growth rate include residual stresses, thermal stresses, metallurgical conditions such as hydrogen in steel and carbide precipitation in alloy steels, and environmental conditions such as corrosive mediums.

Failure stage. When the size of a crack exceeds a certain size, the structure will no longer be able to support the applied loads and will consequently fail. Once the crack reaches its critical length, the failure process is rapid.

The most common cases of mechanically generated cracks are presented in more detail in the succeeding text.

Concrete structures are particularly susceptible to cracks from external loadings because they usually start developing small structural cracks at loads well below service level [6]. Two main types of structural cracks are flexural cracks and shear cracks, schematically shown in Figure 1. Flexural cracks are frequently transversal and found in locations subjected to large moments. On the other hand, shear cracks occur in areas with dominant shear and are identified by their slant (diagonal) direction [6].

Figure 1. Left: a flexural (transverse) crack in concrete; right: a shear (slant) crack in a concrete beam.

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While flexural failures are usually ductile because of the presence of steel reinforcement, shear failures tend to occur suddenly with no advance warning [6]. Therefore, shear cracks are generally considered more detrimental for reinforced concrete structures.

Concrete structures can also develop fatigue cracks. While the fatigue mechanism in concrete is not clearly understood because of the material's heterogeneous composition, studies have shown that fatigue loading on concrete initiates the growth of microcracks at the locations of internal flaws [7,8]. Fatigue crack propagation in concrete depends on the interaction between microcrack growth, which encourages further cracking, and aggregate crack bridging, which prevents further cracking [8]. Depending on the stress magnitude and direction, concrete structures can take more than one million cycles to fail. However, fatigue cracks can cause excessive deformations, large crack widths, and reinforcement debonding—all of which can lead to severe structural failures [9].

Steel structures can exhibit cracking if a critical level of loading or the material's yield strength is reached. Depending on material composition and treatment, the yield stresses of structural steel range from 220 MPa (32 ksi) for carbon steel up to 690 MPa (100 ksi) for quenched and tempered steel [10]. Steel structures may develop brittle and/or ductile cracks if they undergo buckling or torsion from external loads. These cracks develop at locations of maximum tension and compression and propagate perpendicular to the direction of the stresses [11]. Ductile cracks exhibit plastic deformation at their edges and are formed by the coalescence of microvoids that develop from inclusions [11]. Gradual failure occurs when the voids reach a critical size and form a central crack. In contrast, brittle cracks in steel show very little plastic deformation and develop similarly to concrete cracks [11]. Because brittle failure occurs suddenly and rapidly, it is therefore more critical than ductile failure.

Fatigue cracks (Figure 2) occur in steel because of repeated loading cycles. They usually originate on the steel surface because of some discontinuity or stress concentration [11]. After initiation, the crack propagates perpendicular to the direction of maximum tension and compression [12]. The loading opens and blunts the crack tip and the unloading re-sharpens the tip. This repeated process is accompanied by shear, which causes further crack growth [13]. Studies have determined linear and log-linear relationships between the surface crack length and the crack depth [14–17]. While the exact aspect ratio of crack depth to crack length depends on the specific material properties, the shape of a crack front in the interior of the steel normally has a semi-elliptical shape [15]. Fatigue failure accounts for over 80% of all failures in metals because it occurs at stresses well below the yield strength of the material [13,18]. While it takes several years for fatigue cracks to grow, once they reach a critical size the structure will undergo brittle failure with little deformation [11]. Thus, the consequences of fatigue cracks in steel can be severe.

2.3. Cracks caused by physical and chemical processes

In addition to mechanical cracks, concrete and steel can develop cracks due to various physical and chemical processes. These processes, which depend on the particular environmental and chemical properties of the material, typically include the following:

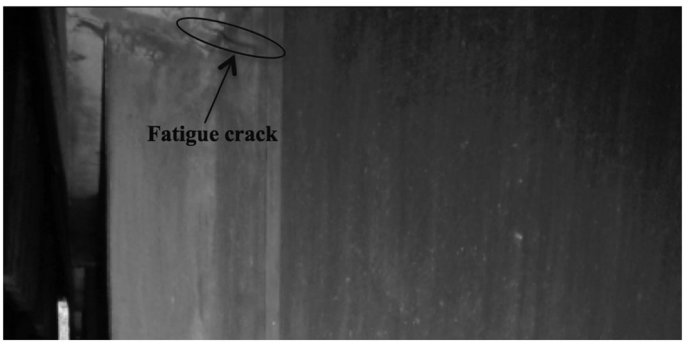

Figure 2. Example of fatigue crack in steel.

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- Cracking in concrete due to

  - early-age deformation (endogenous and plastic shrinkage);
  - drying shrinkage;
  - freeze-thaw cycles;
  - temperature variations and gradients;
  - sulfate attack;
  - alkali–aggregate reaction; and
  - chloride penetration and corrosion in reinforcement bars (rebars)

- Cracking in steel due to stress corrosion.

Shrinkage in hardened concrete can be subdivided into three main categories: drying shrinkage, plastic shrinkage, and endogenous shrinkage [19]. Drying shrinkage strongly depends on the humidity in the environment and the water–cement ratio of the concrete [19]. When wet concrete dries, the evaporation of water develops suction in the pore liquid, which forces the concrete to contract [20]. Early-age plastic shrinkage occurs when the rate of water evaporation from the concrete surface exceeds its bleeding rate, resulting in high capillary stresses near the surface [21]. Endogenous shrinkage is less dependent on the environment and refers to the shrinkage caused by chemical processes (carbonation and hydration) and thermal activity (heat of hydration dissipation) in the concrete [19]. For all cases, shrinkage is greatest at the surfaces of the structure. This causes tensile stresses that can exceed the concrete's tensile strength and develop into surface cracks [19]. Shrinkage cracks are very fine, rarely exceeding 10 mm (0.4 in) in depth and 10 μm (0.0004 in) in width [20]. They are of varying lengths and are generally spaced from a few centimeters up to 3 m (10 ft) apart [22]. In general, shrinkage cracks can accelerate the degradation process of concrete by encouraging waterproofing, chemical leaching, and corrosion problems [23]. An example of shrinkage crack is shown in Figure 3.

Freeze-thaw concrete cracking occurs at locations prone to water saturation and freezing [24]. Above a critical degree of saturation (91.7% of water), ice crystals in concrete will exert a hydraulic pressure inside the pores [25]. If this pressure exceeds the tensile strength of the surrounding material, the structure will develop subsurface cracks that propagate to the surface. Successive freeze-thaw cycles can cause not only cracks, but also pop-outs. A pop-out appears as a circular crack and leaves a crater on the concrete surface; their dimensions range from a few millimeters to 100 mm (4 in) in diameter and up to 40 mm (1.6 in) in depth [20]. Freeze-thaw damage typically takes several years to appear on the concrete surface so they are not as critical as other types of cracks. However, if untreated, they can eventually lead to the total disintegration of concrete [24]. An example of a freeze-thaw crack is given in Figure 4.

Concrete can undergo chemical attacks when certain chemical solutions leach into the structure. The presence of sulfate facilitates the expansive formation of ettringite or gypsum inside the concrete. Because the chemical products are more voluminous than the surrounding cement, they will induce an internal pressure that may be sufficiently high enough to cause parallel subsurface cracking [20]. Extensive cracking inside the structure will eventually lead to the exfoliation and delamination of concrete surfaces. Additionally, alkali–silica reactions occur when the cement mixture is too alkaline. Silicate ions in the aggregate can react and form alkali-silicate gel, a product that is also more voluminous than concrete [20]. Isolated cracks form during the early stages of alkali–silicate reaction, terminating at depths of 50–75 mm (2–3 in). However, the cracks form joined networks as the reaction develops, as shown in Figure 4. This may eventually lead to exfoliation and pop-outs [20]. While they

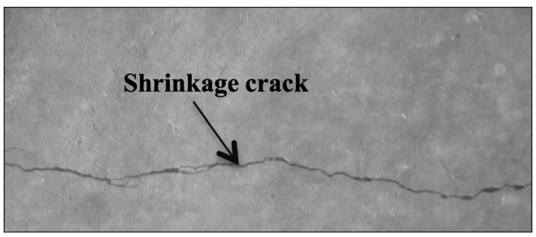

Figure 3. Shrinkage crack.

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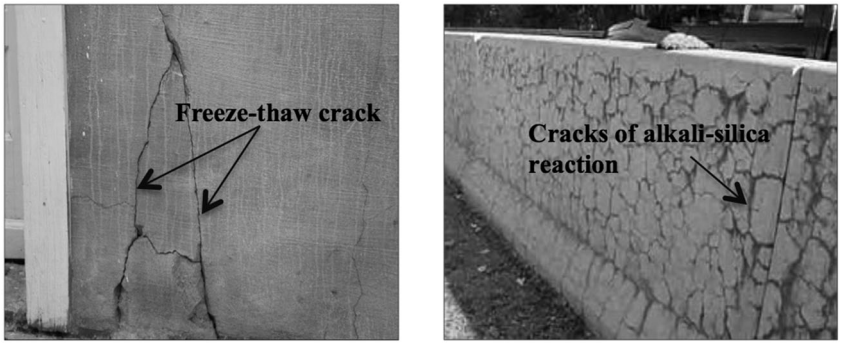

Figure 4. Left: freeze-thaw crack (source: inspectapedia.com, photo: Daniel Friedman [109]); right: cracks formed by alkali–silica reaction (source: wikipedia.org, photo: Dr. P.E. Grattan-Bellew, Federal Highway Administration [110]).

may take several years to develop, chemical cracks eventually lead to material loss, which severely reduces the structure's capacity and increases its vulnerability to further chemical attacks.

Corrosion cracks can occur in reinforced concrete because of rebar corrosion. When the pH of the pore solution in concrete is high, a thin oxide layer forms on the surface of reinforcement [20]. If exposed to low-pH moisture, the protective oxide layer will break down and the steel will react with water, forming rust. This typically happens when chloride penetration (e.g., dissolved de-icing salt) reaches the rebars. Because the volume of reaction products is greater than the initial volume of rebars, an expansive pressure will form inside the concrete and cause rupture [26]. The resulting cracks begin adjacent to the rebars as subsurface cracks and can propagate to the surface. While they do not have urgent consequences when they are internal, rebar corrosion cracks become extremely critical once they cause the surrounding concrete to spall as this reduces the cross-section area of structural members and further expose rebars to accelerated corrosion [20]. An example of a corrosion crack is shown in Figure 5.

Temperature gradients and differential thermal expansions can have damaging effects on structures. Globally, temperature variations may generate stresses that facilitate mechanical cracking in statically indeterminate structures. Locally, they can change the mechanical properties of the material and lead to its deterioration, especially in the event of fire. Structural steel elements are usually fireproofed, and therefore, thermal cracking is more prevalent in concrete. Initial heating evaporates the pore water and actually causes cement paste contraction. However, the aggregate undergoes progressive expansion, opposing the shrinkage and possibly rupturing the surrounding cement [20]. Fire damage can cause random cracking networks and surface spalling, and the latter is more critical because it is explosive and unpredictable [20]. Heating below 250 °C frequently does not cause stresses large enough to cause significant cracking [20].

Steel elements may undergo stress corrosion cracking (SCC) when they are simultaneously exposed to tensile stresses and corrosive environments such as chlorides (Figure 6). In general, low carbon steels become more susceptible to SCC as the carbon content increases [11]. Studies have shown that there is a threshold tensile stress intensity, below which SCC does not occur. This stress intensity varies depending on the metal, heat treatment, and surrounding environment [11]. SCC cracks usually form at physical discontinuities due to chemical accumulations and propagate along the surfaces of bends and welds due to high residual stresses. Initially, the cracks are very fine and penetrate deeply

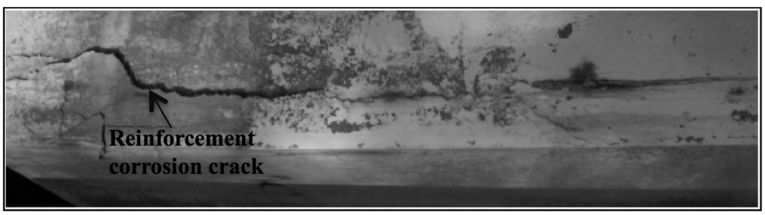

Figure 5. Reinforcement corrosion crack.

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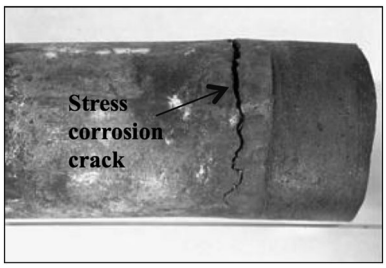

Figure 6. Stress corrosion cracking in steel (source: Metallurgical Technologies, Inc. [111]).

into the material without obvious signs of deterioration. However, once the cracks grow to a critical size, brittle fracture can suddenly occur [11].

2.4. Summary on crack classification, causes, and consequences

The various types of cracks found in the two most common materials in civil engineering practice are described in this section and concisely presented in Tables I and II. In summary, cracks have adverse effects on structures; initial cracks may be caused by a single mechanism, but once it has been created, it can trigger or accelerate other processes that could cause crack propagation. For example, a mechanical crack may act as a channel for accelerated penetration of chemicals, which can then in turn facilitate rebar corrosion or alkali–silicate reaction. The delamination of concrete due to rebar corrosion reduces the cross-section capacity of the structure, which can lead to further crack initiation or growth. Consequently, all cracks can potentially cause a dramatic decrease in structural capacity and durability. It is therefore important to detect cracks during their early stages so that any maintenance and preservation work can be planned timely and efficiently.

3. CRACK DETECTION AND CHARACTERIZATION TECHNIQUES BASED ON DIRECT SENSING

Sensor-based crack detection has been proven effective when the crack occurs close to or in direct contact with the sensors (e.g., [27,28]) or sensing medium (e.g., elastic waves). The crack is detected and localized directly as an unusual change in the output from the sensors affected by the crack. Hence, this approach in crack detection and localization is called direct sensing in this paper. Typical examples of technologies that enable direct sensing include discrete strain sensors (electrical or fiber optic, short-gage or long-gage), distributed strain sensors (electrical or fiber optic), wave monitoring sensors (acoustic emission and wave propagation sensors), and eddy current sensors. Discrete and distributed strain sensors have to be in direct contact with the potential cracks in order to detect them, while wave monitoring and eddy current sensors can detect cracks as a perturbation in the elastic wave or electric field patterns. These techniques are described as follows.

3.1. Discrete strain monitoring

A material fails when the stress at a certain point exceeds the strength of material. Strain is a parameter directly correlated to stress. There is no effective means to monitor stress under real, on-site conditions; consequently, strain is one of the most relevant parameters to monitor [29]. Electric and fiber-optic sensors are the two most used strain-monitoring technologies. Each technology features discrete and distributed sensors.

Resistive strain gages (see examples in Figure 7) are the most mature and widely used sensors for strain monitoring. They are commercially available in many forms, with gage lengths ranging between 0.4 and 100 mm (0.016 and 4 in). The strain gages are usually bonded to the surface of the structure by suitable adhesives. A strain change in the structure would be transmitted to sensor, which could then be accurately measured by an appropriate reading unit. Strain gages have a long history in crack detection. For example, Heise [30] presented a laboratory test on a steel box-like structure in which the strain

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Table I. Steel crack summary.

	Mechanical causes		Chemical causes
	Loading	Fatigue	Stress corrosion
Causes	Loading stresses that exceed the yield strength	Repeated loading cycles (especially at locations of discontinuities or stress concentrations)	Simultaneous exposure to tensile stresses and corrosive environments
Physical appearance	Perpendicular to the direction of maximum principal stress	Perpendicular to the direction of maximum principal stress	Form at physical discontinuities and propagate along the surfaces featuring high residual stresses
Size and shape	Crack size and shape vary depending on loading	Initial cracks are minute. Lengths and depths are correlated; crack fronts are usually semi-elliptical	Initial cracks are minute but penetrate deeply into the material
Consequences	Ductile failure is gradual. Brittle failure is sudden and rapid	Brittle failure at stresses well below the yield strength	Brittle failure (sudden and rapid)

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Table II. Reinforced concrete crack summary.

	Mechanical causes	Physical and chemical causes	
	Loading and fatigue	Shrinkage[1], freeze-thaw[2], and thermal[3]	Chemical reactions and corrosion
Causes	Bending moments cause flexural (transverse) cracks Shear stresses cause shear (slant) cracks Repeated loading cycles cause fatigue cracks	[1]Evaporation causes drying and plastic shrinkage, hydration and carbonation cause endogenous shrinkage [2]Water saturation (>91.7%) and freezing temperatures [3]Temperature gradients and differential material expansions	Leaching of sulfate solutions and the presence of contaminated aggregate prone to alkali–silicate reaction Reinforcement bars exposed to penetrated chlorides corrode and expand
Physical appearance	Perpendicular to the direction of maximum principal stress	[1]Shrinkage causes fine surface cracks [2]Freeze-thaw causes surface cracks and pop-outs [3]Temperature causes surface crack networks and spalling	Alkali–silicate reaction causes surface crack networks, exfoliation, and pop-outs Rebar corrosion causes subsurface cracks and spalling.
Size and shape	Loading crack widths vary depending on loading type Fatigue cracks are initially minute, but developed cracks may have large widths	[1]Shrinkage cracks vary in length but rarely exceed 10 mm in depth and 10 μm in width [2]Size of freeze-thaw surface cracks vary; pop-outs are usually <100 mm in diameter [3]Thermal cracks vary in size	Initial alkali–silicate reaction cracks are typically 50–75 mm deep Size and depth of corrosion cracks can vary (depending on the position of rebars and concrete layer below surface)
Consequences	Flexural failures are gradual Shear failures are sudden Fatigue cracks may lead to excessive deformations, reinforcement debonding, and brittle failure	[1,2]Shrinkage and freeze-thaw cracks facilitate further deterioration from freeze-thaw cycles and chemical attacks [3]In addition, thermal cracks may cause concrete loss and a significant reduction in strength capacity	Loss of concrete and significant reduction in strength capacity

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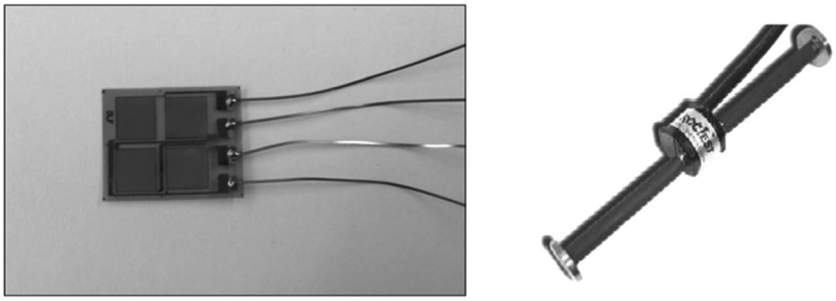

Figure 7. Left: an example of resistive strain gage; right: an example of a strain sensor based on vibrating wires (courtesy of Roctest, Inc.); images are not scaled.

gage was able to follow the initiation and propagation of cracks caused by low-cycle fatigue. This early test not only proves the sensitivity and accuracy of strain gages in crack detection but also shows the general ability of strain sensors to detect and characterize cracks. Besides resistive strain gages, other widely used electrical discrete strain sensors are those based on vibrating wires (Figure 7). Sensors based on vibrating wires have the advantage of incorporated temperature compensation in the sensor packaging. They also feature longer gage lengths, typically ranging between 50 and 150 mm (2 and 6 in), and can be both embedded in concrete or mounted on the surface of steel and concrete structures. The longer gage length makes them more sensitive to crack occurrences because the increased contact length with the structure increases the probability of the sensor to capture the crack. However, both resistive strain gages and wire vibrating sensors can be considered as short-gage sensors.

Discrete strain sensors based on fiber-optic technologies have reached market maturity during the last decade. Their functioning and performance were proven not only in harsh conditions frequently found on-site [31] but also in very extreme environments [32]. While there are several physical principles for discrete sensors, the most widely used sensors are those based on Extrinsic Fabry–Perot Interferometry (EFPI), fiber Bragg grating (FBG), and low-coherence interferometry (SOFO). EFPI sensors have gage lengths ranging between 51 and 70 mm (2 and 2.75 in), while FBG sensors can be made with a wider range of gage lengths from 10 mm (4 in) up to 2 m (6 ft 6.7 in). SOFO sensors are exclusively long-gage sensors with lengths ranging between 250 mm (10 in) up to 20 m (66 ft). More information on the functional principles of these sensors can be found in literature [31]. Several field application examples of the aforementioned sensors are also presented in literature [27,31,33,34], including crack detection in bridge deck, tunnel vaults, and pile foundations. The robustness of these sensors in damage detection is based on the direct detection principle and outlier identification algorithm in which the signal generated by damage dominates the environmental noise (see further text). Compared with electric strain sensors, short-gage fiber-optic sensors such as EFPI and FBG (Figure 8) are more sensitive and durable (they are insensitive to electromagnetic influences, humidity, and

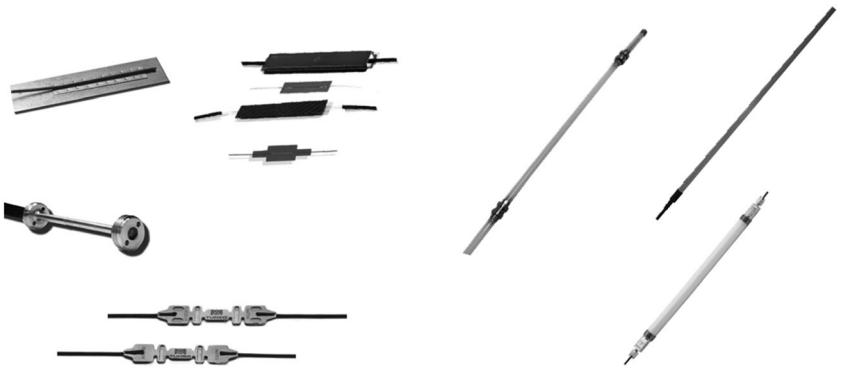

Figure 8. Examples of fiber-optic strain sensors; left: short-gage sensors (courtesy of Roctest, Inc., Fibersensing, and Micron Optics); right: long-gage sensors (courtesy of Micron Optics and SMARTEC); images are not scaled.

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corrosion) but have substantially higher cost. Similar to vibrating wire sensors, short-gage fiber-optic sensors can be both embedded in concrete or mounted on the surface of steel and concrete structures.

The key challenges of short-gage sensors (electrical or fiber optic) are (1) their insensitivity to strain-field anomalies that do not occur in the immediate proximity of the sensor [35] and (2) the large number of sensors required to cover a large area, thereby imposing complex cabling, deployment, and data acquisition (e.g., [30]).

Long-gage fiber-optic strain sensors (e.g., [27,31], Figure 8) enable larger coverage for crack detection than short-gage sensors because of their longer gage length. Several examples from the past have shown successful crack detection and characterization (localization and quantification) by long-gage sensors (e.g., [27,36,37]). Two examples of field applications are given in Figure 9 [27,36]: the crack crosses the locations of sensors and generates a significant strain increase for both sensors, revealing that a crack has occurred and indicating location and the size of the crack (more details are given in [36,27]).

Figure 9 shows that direct sensing provides very reliable crack detection, given that the crack is in contact with the sensor (e.g., crack detection would be successful even if the noise generated by the variable environmental conditions and the monitoring system were an order of magnitude larger). However, the reliability of detecting and characterizing damage that occurs in locations far from the sensors is challenging. In some cases, this problem can be solved by using sophisticated algorithms (indirect sensing, see the next section). However, the performance of such algorithms may be challenged in on-site conditions because of various influences that may 'mask' the damage such as high temperature variations and load changes, and outliers and missing data in monitoring results [38]. Although long-gage sensors have improved capabilities for crack detection and characterization when compared with short-gage sensors, they can also suffer from insufficient spatial sensor coverage. Figure 10 compares damage detection capabilities of short-gage, long-gage, and distributed sensors on the basis of their spatial coverage in the structure.

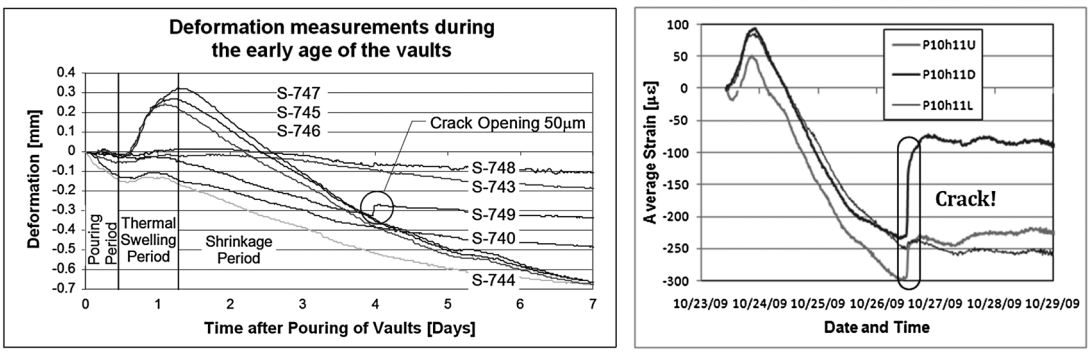
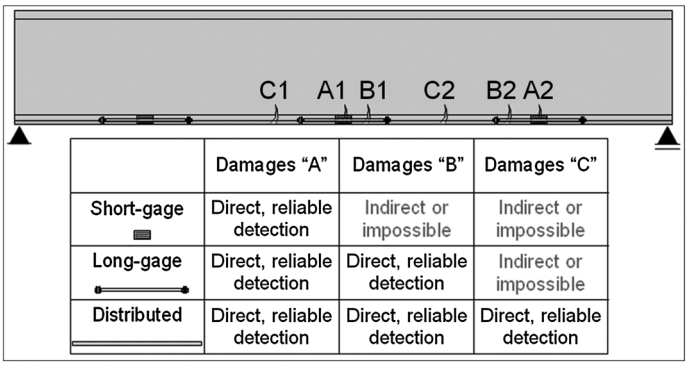


Figure 9. Examples of crack detection and localization in real structures using long-gage fiber-optic strain sensors; left: in a tunnel [36]; right: in a bridge [27].



	Damages "A"	Damages "B"	Damages "C"
Short-gage	Direct, reliable detection	Indirect or impossible	Indirect or impossible
Long-gage	Direct, reliable detection	Direct, reliable detection	Indirect or impossible
Distributed	Direct, reliable detection	Direct, reliable detection	Direct, reliable detection

Figure 10. Schematic comparison between damage detection capabilities of short-gage, long-gage, and distributed sensors.

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The discrete strain sensors (short-gage or long-gage) are usually placed at locations where high strain is expected and the crack is more likely to occur (e.g., damage 'A' is expected in the middle of a simply supported beam; see Figure 10). However, because of local variability in material properties and dimensions, the cracks may not necessarily occur exactly at the locations of the highest stress but at locations that are relatively distant from these locations (e.g., damage 'B' or damage 'C' occurs instead of damage 'A' in Figure 10). Short-gage sensors would not be able to directly detect the damage in that case as they are not in direct contact with the damage. The long-gage sensors extend over larger areas, and thus, they are more likely to directly detect the damage that does not occur exactly at the expected locations (i.e., they can directly detect damage 'B'), but they are still inefficient if the damage does not occur at the location of the sensor (damage 'C').

In conclusion, discrete sensors (electric or fiber optic, short-gage or long-gage) that are sparsely spaced in real structures may suffer from insufficient spatial coverage. However, if the approximate locations of crack occurrences are known (e.g., on the basis of numerical modeling), then long-gage sensors can provide much better crack detection capabilities than short-gage sensors. Increasing the number of sensors may help as well but only to a certain extent because this would involve increased hardware and installation cost.

3.2. Distributed strain monitoring

Distributed strain sensing can overcome the spatial coverage and specificity challenges related to discrete sensing. A distributed sensor can be represented by a single cable that is sensitive at every point along its length. Hence, one distributed sensor can replace a large number of discrete sensors. Moreover, it requires a single connection cable to transmit information to the reading unit instead of using a large number of connecting cables as in the case of wired discrete sensors. Finally, distributed sensors are less difficult and more economical to install and operate on large structures. An illustrative comparison between a pipeline structure equipped with distributed and discrete sensors is shown in Figure 11 [28].

Coaxial cable sensors [1], as shown in Figure 12, are distributed sensors with a high potential for detecting cracks and measuring strain in reinforced concrete structures. They are based on electrical time-domain reflectometry (ETDR). The components of coaxial cable sensors are shown in Figure 12 [1]. The inner conductor is separated from a steel spiral by using Teflon, and the assembly is covered with a thin layer of solder to create a continuous outer conductor [39]. The sensor is designed to be embedded in concrete. Cracks developed in concrete structures can lead to the separation of the local steel spiral, and this large discontinuity can be then detected with an ETDR reading unit. As a result, a cable sensor in the concrete can detect both the location and width of cracks by calculating the reflection coefficient, which indicates the intensity of reflected waves through multiple cracks [1]. Chen *et al.* [5] performed both laboratory testing and field application of crack detection in a reinforced concrete bridge in Missouri, and they concluded that the measured reflection coefficient correlates well with

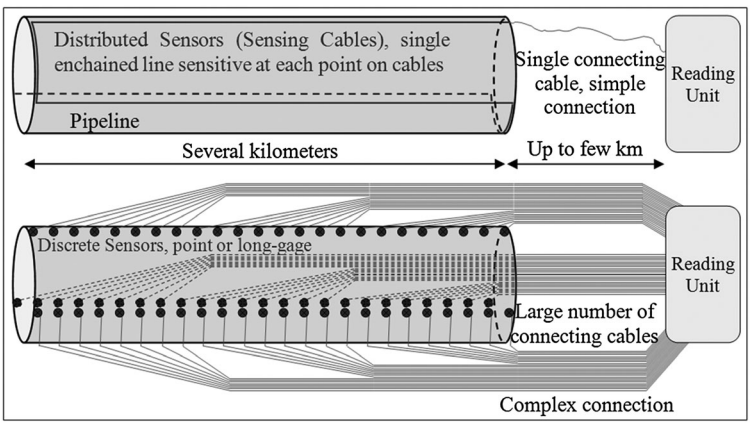

Figure 11. Distributed versus discrete monitoring [28].

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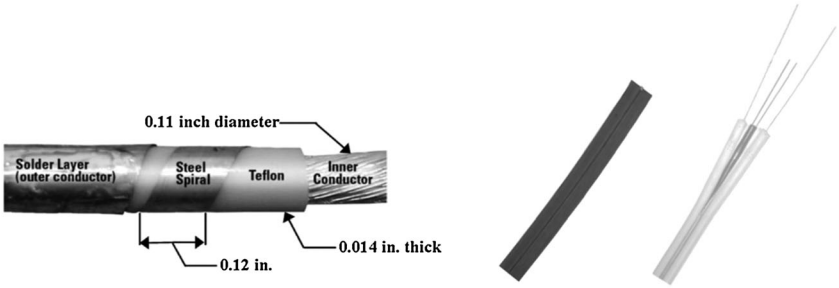

Figure 12. Distributed sensors; left: example of coaxial cable [5]; right: examples of fiber optic distributed sensors (courtesy of SMARTEC); images are not scaled.

crack width. The major challenge for this sensing technique is related to signal losses over large distances—the site application was based on a 2.3-m-long (90 in) sensor, and the longest measurable length of the sensor was not documented. Other challenges include the application of the sensor to existing concrete and steel structures and repair and replacement of the malfunctioning sensor.

There are two main principles for distributed strain sensing in the domain of Fiber optic sensors (FOS) the Rayleigh scattering effect (e.g., [40]) and the Brillouin scattering effect (e.g., [41]), which can be stimulated (BOTDA) or spontaneous (BOTDR). Each technique is based on the relation between the measurand (i.e., strain) and the encoding parameter (i.e., the change in optical properties of the scattered light). Rayleigh-based sensors feature better spatial resolution (10 mm) than Brilouin-based sensors (500–5000 mm), but they are limited to modest lengths of 70 m. Stimulated Brillouin scattering is the least sensitive to optical losses and allows long distance monitoring of up to 400 m without channel switches. With channel switches, the monitoring distance is practically unlimited [42] (an application involving application of 5 km on a real steel bridge is given in [41]). The sensors can be both installed on the surface of existing structures (e.g., [28,41]) and also embedded in the concrete of new structures (e.g., [27]).

Distributed fiber-optic sensing systems based on Brillouin scattering were designed for detecting cracks with widths larger than 0.5 mm [43]. Examples of crack detection in real on-site conditions and close-to-real conditions are given in Figure 13.

Similar to discrete sensors, the crack is detected directly as an unusually high strain change. While the localization of the crack is very simple (position of strain change along the cable), the size of the crack is challenging to determine because of the averaging of the measurements and the accuracy of the sensing system. Figure 13 (right) shows the comparison between distributed and long-gage sensors at the location of the crack opening. The discrete sensors feature better measurement properties (higher resolution, accuracy, and frequency of reading). However, distributed sensors feature better spatial coverage.

The packaging of optical fibers in a sensing cable and the surface installation remain as challenges for distributed fiber-optic sensors, as the former issue introduces optical losses that result in shorter

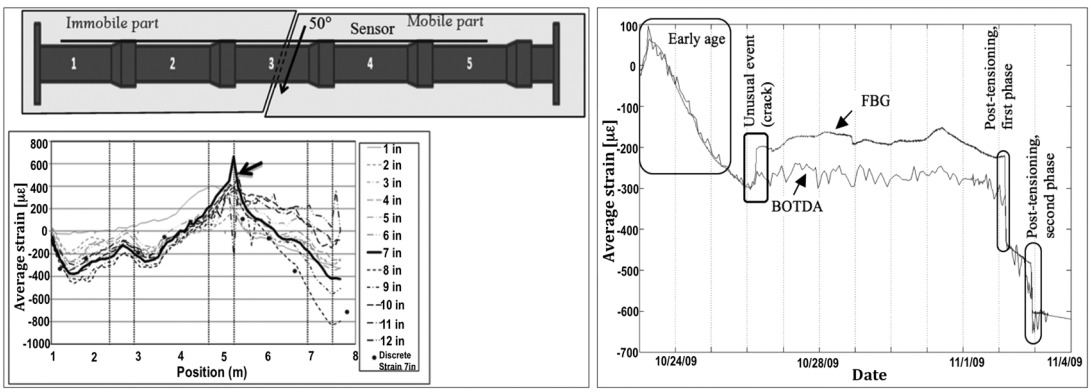

Figure 13. Crack detection and localization using distributed fiber-optic sensors; left: on a large-scale test of pipeline exposed to permanent ground movement [28]; right: on a pedestrian bridge due to thermal gradients at early age of concrete [27].

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monitored lengths, and the latter issue is time consuming and cost sensitive. Another important technical challenge for distributed fiber-optic sensors is repair and replacement of malfunctioning sensor.

3.3. Direct sensing based on wave monitoring

Acoustic emissions occur when discontinuities in the structure such as crack openings release energy. This energy travels in the form of high-frequency acoustic (elastic) waves, which can provide comprehensive information about the origination of a discontinuity (crack) in a stressed component and the development of this discontinuity (e.g., change in size) caused by continuous or repetitive stresses. Because acoustic waves travel through the structure, it is possible to localize the position of a crack by placing various transducers over the structure and determining the wave speed and the propagation time. Even relatively weak acoustic emissions can be used for crack detection in metallic structures [44]. Grosse [45] used acoustic transducer network to detect the damage on a prestressed reinforced concrete bridge built in a large test facility in Germany. The transducers were optimized to operate on signals up to 25 kHz, and they successfully detected and characterized (localized and quantified) the crack with the signal processing techniques of beamforming and velocity spectral analysis. An example of the acoustic transducer is shown in Figure 14. The challenge of this method is in the creation of complex algorithms that have to guarantee robustness in real on-site conditions, where many sources of noise are present. Another challenge is the detection of cracks that may occur while the monitoring system is off.

Piezoelectric transducers (e.g., [46]) can be used as both actuators that generate elastic waves and detectors of the waves incoming from other piezoelectric elements. Typically, they can introduce elastic waves into the structural member. The propagation properties of waves (such as attenuation, velocity, or reflections) can be correlated with the health condition of the member, and if a crack is present, it can be detected and characterized. There are several types of elastic waves that can be used for damage detection and characterization purposes, and the most frequently used ones are briefly presented here. *P*-waves are longitudinal waves propagating through continuous materials. They are commonly used for the NDE of concrete [47]. Rayleigh waves propagate near the surface of solids, and they are used to detect defects in steel and concrete [48]. Lamb wave is an elastic wave propagating along a thin solid (plate), whose particle motion lies in the plane defined with the direction of wave propagation and the direction normal to the surface of the thin propagating solid (plate). In steel structures, the Lamb waves can be directed [49,50] and made to propagate over large distances (e.g., in pipelines up to 1 km). Figure 14 shows an example of the piezoelectric transducer.

By using active sensors (that serve as wave emitters and receivers) and performing appropriate data analyses, it is possible to detect and localize cracks. The transducers must be carefully bonded to the structure, and emitter/receiver distance limitations imply the need for a large number of sensors and cables to cover large areas of the structure. This significantly increases the cost of monitoring. The data required from all of the sensor channels can also become extremely complex to analyze.

Eddy-current-based crack detection techniques exploit the fact that damage (e.g. cracking) alters the flow of the eddy current generated in the structure, and this is detected as a variation in the remote magnetic field (Figure 15 [51]). The technique is mostly applied on metallic structures that are good electrical conductors. A driving coil is used to generate the eddy current in the structure while the sensing coil

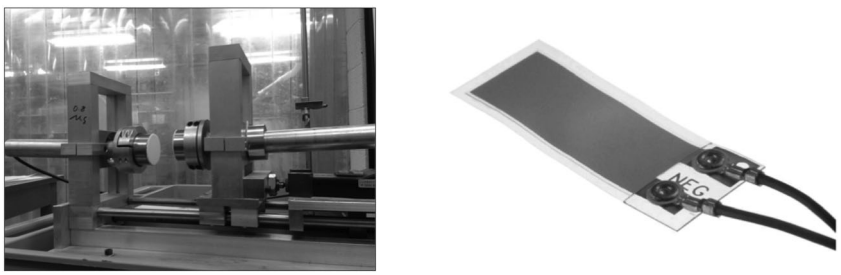

Figure 14. Left: acoustic transducer; right: piezoelectric transducer (source: model LDT1-028 K, Measurement Specialties, Inc. [112]); images are not scaled.

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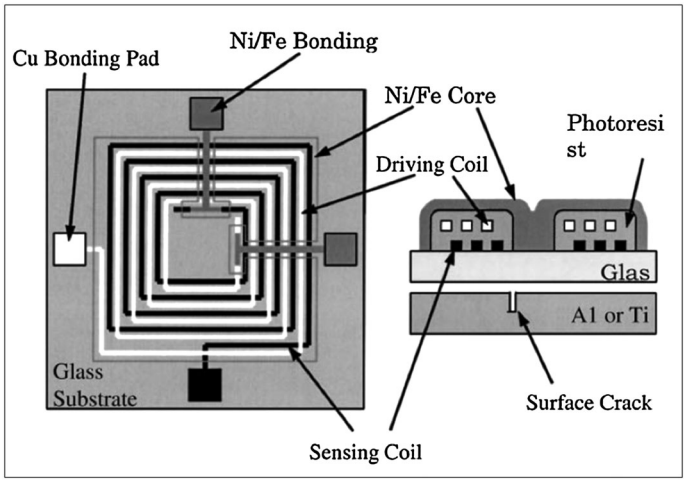

Figure 15. Top view (left) and side view (right) of the eddy current sensor [51].

is used to detect anomalies in remote magnetic field. Sadler and Ahn [51] presented an integrated eddy current sensor for proximity sensing and detecting microcracks on metallic surfaces. They demonstrated that their sensor is capable of detecting crack depths that are as small as 200 μm. Najafi [52] presented the use of eddy-current-based techniques for damage detection in pipelines. The technique is typically used in NDE as a movable proximity sensor and therefore does not allow for continuous monitoring unless numerous transducers are installed on the structure, which is similar to the challenges in acoustic emission and wave propagation techniques.

3.4. Summary on direct sensing techniques

In summary, direct sensing techniques are successful in crack detection and characterization. Although they have various advantages, they also face some challenges. Both advantages and challenges are presented in Table III.

4. CRACK MONITORING TECHNIQUES BASED ON INDIRECT SENSING (MODELS AND ALGORITHMS)

4.1. General observations

Indirect sensing methods are based on measurements made by sensors that are not necessarily in direct contact with damage (e.g., accelerometers, strain sensors that are not at the location of the damage, remote sensors, etc.). As opposed to direct sensing where the damage is detected directly as a noticeable change in output signals of affected sensors, in the case of indirect sensing, the sensors usually do not manifest noticeable changes in output signals. Consequently, the recorded data have to be analyzed using various classes of sophisticated algorithms in order to ascertain crack detection and perform crack characterization. The deployed sensing technique can be based on various types of sensors such as strain sensors (not in contact with the damage), remote sensors (scanning laser vibrometer), and so on. However, the techniques that are frequently applied at present are related to measurements performed with accelerometers. This overview mostly (but not only) focuses on these techniques. Generally, two approaches in data analysis are identified: model-based and model-free (data-driven) approaches.

4.2. Model-based approaches

Model-based algorithms use mathematical models describing structural behavior to infer crack occurrences, locations, and sizes. A typical example is modal analysis, where the data collected by

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Table III. Comparison of various direct sensing techniques currently used for crack detection and characterization.

	Discrete strain monitoring				Distributed strain monitoring		Wave-based monitoring		
Type of sensor	Resistive strain gage	Electrical vibrating wires	FOS short gage	FOS long gage	Coaxial cable	FOS distributed	Acoustic	Piezoelectric	Eddy current
Function principle	Electric resistance change	Vibrating wires	EFPI, FBG	FBG, SOFO	ETDR	Rayleigh and Brillouin Scattering	Acoustic emission	Wave propagation	Eddy current
Structural material and type	All, new and existing structures (surface mounted)	All, new and existing structures	All, new and existing structures	All, new and existing structures	Concrete, new structures	All, new and existing structures	All, new and existing structures	All, new and existing structures	Metal, new and existing structures
Application guidance	Local monitoring, more useful if crack location can be predicted; short gage mostly for homogeneous materials (e.g., metals and composite materials in the direction of reinforcing fibers), long gage for all materials				Global monitoring, more useful if crack location cannot be predicted (all materials); acoustic and piezoelectric perform better on metals and composite materials but can also be used on concrete				
Gage length/ length range	0.4–100 mm (0.016–4 in)	50–150 mm (2–6 in)	EFPI: 51–70 mm (2–2.75 in) FBG: 0.01–2 m (4–80 in) SOFO: 0.25–20 m (10 in–66 ft)		N/A (Limited due to signal loss)	Rayleigh: 70 m (200 ft) Brillouin: 5 km (3 mi)	Depends on material properties, wave frequency, and environmental noise; typical 0.2–70 m (8 in–200 ft)		0.1–1 m (4–40 in)
Advantage	Low cost, high accuracy, simple data processing	High stability and accuracy, simple data processing	High durability, stability, and accuracy, simple data processing	Improved spatial resolution, durability, stability, and accuracy, simple data processing	High spatial resolution	High spatial resolution, durability, and stability, simple data processing	High spatial resolution and sensitivity		High sensitivity
Challenge	Low spatial resolution, complex cabling, long-term drift, affected by electromagnetic interference	Low spatial resolution, complex cabling, affected by electromagnetic interference	High cost, low spatial resolution	High cost, limited spatial resolution	Signal loss, short-length range, surface installation, sensor repair	Rayleigh: short-length range Both: sensor packaging, complicated surface installation, sensor repair	Complex cabling, affected by environmental noise, complex data analysis		Real-time damage detection (mostly used as NDT and not SHM)

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accelerometers are processed to calculate the natural frequency, mode shapes, damping, and response to specifically applied excitation. Changes in these parameters are then associated with cracking through various algorithms.

Bachschmid *et al*. [53,54] presented a combined model-based method for crack localization and depth determination. The method was verified in a laboratory test on a steel beam. The crack position is determined by using a model-based diagnostic approach and a least-squared identification in frequency domain. The crack depth is determined by comparing the static bending moment generated by self-weight to identify 'equivalent' bending moments that simulate cracks. Ray and Kishen [55] introduced an analytical model to estimate the fatigue crack growth in concrete. The fatigue crack growth rate depends on several parameters, for example, tensile strength, fracture toughness, loading ratio, structural size, and so on. This model considers the mentioned parameters and combines them with the concepts of dimensional analysis. They concluded that this approach is able to capture the size effect in concrete and that laboratory experimental results supported this method. Ray and Kishen [55] also found that structural size plays a more important role in determining the fatigue crack propagation than other factors such as loading ratio and initial crack length.

Friswell and Penny [56] made a review paper about crack modeling in beam structures. They summarize all crack modeling approaches into three main categories: local stiffness reduction, discrete spring models, and complex models (in two or three dimensions). After numerically comparing different methods, they concluded that for low-frequency vibration, local stiffness reduction approaches are adequate to model the cracks. In addition, they discussed the nonlinear dynamics of a breathing crack and indicated that with a bilinear stiffness model for crack opening and closing, the impulse and random responses of the beam approximate a linear response with natural frequencies. They were able to accurately detect and localize the cracks in a small beam tested in a laboratory, while the crack depth estimation would be smaller than reality. Kim and Stubbs [57] presented crack localization and quantification models for beam structures. The models are based on the relationship between fractional changes in modal energy and changes in a few natural frequencies caused by cracks. The natural frequencies of the tested beams are determined and compared for different crack locations and sizes. The laboratory tests confirmed that both the crack location and crack size could be estimated with a small error. Wang and Qiao [58] improved the method by introducing the uniform load surface (ULS) technique, which combines generalized fractal dimension (GFD) and simplified gapped-smoothing (SGS) methods. The peak on the GFD curve of ULS indicates both the position and size of the damage, and the SGS method uses the simple deformation shape of ULS for damage detection. Both methods are verified through comparisons with laboratory tests involving cracked and delaminated composite beams.

Dimarogonas and Papadopoulos [59,60] and Papadopoulos and Dimarogonas [61,62] presented a method based on a flexibility matrix they developed to identify transverse cracks and the influences of coupling between axial and torsional vibrations and bending on cracks in a beam structure. Miller *et al*. [63,64], Brook *et al*. [65], Rajab *et al*. [66], and Tsai and Wang [67] also presented methods for crack localization and quantification based on the analysis of changes in natural frequencies and mode shapes.

Pai *et al*. [68] introduced a boundary effect evaluation (BEE) method for localizing cracks and estimating crack sizes that utilizes operational deflection shapes (ODS) that are measured by a scanning laser vibrometer. The BEE method extracts boundary solutions from experimental ODS, and cracks are localized where crack-induced boundary solutions are different from those of actual boundaries. A local strain energy method is used to estimate the crack size through experimental ODS and stress intensity factors (from fracture mechanics). Laboratory experiments indicate that the BEE method can be used to localize and quantify small cracks.

In spite of their abilities to detect and characterize the cracks in laboratory conditions, the aforementioned methods face several challenges when applied in real-life settings: (1) modal parameters are not sensitive to very small changes in the strain field when caused by minute cracks (e.g., [69]), which limits early crack detection; (2) the precise location of the crack is difficult to determine [70] because all methods suffer from cumulated errors generated by complex data processing; and (3) typical noise caused by the normal use of the structure (e.g., traffic on the bridge) and environmental factors (e.g., temperature and humidity variations) may influence the measurements, especially those related

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to modal parameters (e.g., [70]). This may result in ambiguous data interpretation. Consequently, the reliability of crack detection and the accuracy of crack characterization from indirect model-based approaches are challenged in uncontrolled outdoor conditions, that is, in real-life settings.

4.3. Model-free approaches

Model-free approaches are based on statistical data analysis, which is applied to monitoring data assuming that a mathematical model of the structural behavior does not exist. These methods involve the use of sophisticated algorithms such as neural networks [71], fuzzy logic [72], wavelets [73], pattern recognition [74], robust regression analysis, moving principal components analysis [75], and so on.

Adewusi and Al-Bedoor [76] presented a neural network method for structural crack detection. In their laboratory experiments, signals of vibration frequencies with and without propagating cracks were selected to train multilayer feed-forward neutral networks. Other sets of data were tested with the trained neural networks. They concluded that a simple two-layer feed-forward neural network is satisfactory in detecting propagating cracks, while trained three-layer networks are successful in detecting both propagating and non-propagating cracks. Yin *et al*. [77] proposed a probabilistic approach to find the crack characterization in a thin-plate structure, which was based on Bayesian statistical system identification framework. This method only deals with a few points, and the sensors do not necessarily have to be close to the cracks. The crack characterization is performed on the basis of statistics, from which the uncertainty and confidence level of crack location, length, and depth are determined. The authors verified the feasibility of this probabilistic approach with numerical case studies on a rectangular aluminum plate. Furthermore, Sbarufatti *et al*. [78] made a combination of Bayesian Hierarchical Models and artificial neural networks methods to analyze the problem of crack detection and localization on an aerospace structure. An index output related to crack position and length is obtained on the basis of the noisy signal inputs, and cracks have been accurately identified in the majority of the cases.

Umesha *et al*. [79] presented a method for crack detection based on wavelets. They measured the deflections of a beam at discrete points and obtained the wavelet coefficients (WCs) to calculate the position and depth of the crack. They proposed a parametric study by varying the type of damage, location of damage, intensity of load, flexural rigidity, and length of the beam. The WCs vary because of the changes of such inputs. In conclusion, they presented a generalized curve to quantify the crack location and depth in a fixed beam by utilizing the maximum WCs envelope of the deflection response at the damaged points.

Feldman and Seibold [80] researched an approach that combined model-free and model-based procedures for detecting the size and location of the damage. First, the Hilbert transform was used for signal processing, obtaining a slow-varying signal envelope and a phase angle, and detecting nonlinearities. Then, the Kalman filter was applied for detecting the size and location of the damage on the basis of the model information provided by the Hilbert transform. Rehman *et al*. [81] also presented a method based on the Hilbert transform to detect and classify the nonlinearities in the flexural and torsional vibrations generated by a crack. Crack depths were determined on the basis of resonant frequencies and damping levels that are extracted from amplitude-dependent frequency-response functions.

Vakil-Baghmisheh *et al*. [82] presented genetic algorithms (GAs) for crack detection in beam-like structures. They used GAs to detect the changes in natural frequencies of a cantilever beam. The detection of crack location and depth was transformed into an optimization problem. Both binary and continuous GAs (BGA and CGA) were utilized to solve the problem by minimizing the optimization function, which is related to difference between calculated and measured natural frequencies. They concluded that for BGAs, the average prediction errors of crack location and depth are 1.02% and 1.98%; for CGAs, these errors are 0.73% and 1.11%, respectively. Laboratory experimental results verified the accuracy of this GA method.

The main challenges for the application of model-free approaches are related to its sensitivity to noise generated by loading or environmental variations and the lack of supervised training for algorithms. Both challenges can lead to false positive and negative detections of cracks, which reduces the reliability of such methods in real-life settings.

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The various types of indirect sensing methods discussed earlier, including model-based testing and model-free algorithm analysis, are briefly summarized in Table IV. Although this is not an exhaustively complete list of every method (there are indeed a lot more), this limited selection outlines the diversity of current research activities.

5. EMERGING MONITORING TECHNOLOGIES AND FUTURE TRENDS

Studies on crack detection and characterization actively and rapidly evolved in the last decade, and many novel technologies have emerged with the potential to overcome the limitations of the presented techniques. These emerging technologies include wireless sensors, MEMS, thin-film and piezoelectric paints, nano-technologies, LAE, and noncontact sensing techniques.

Wireless sensors and sensor networks (e.g., [83,84]) are an emerging sensing paradigm with a great benefit: they significantly reduce the cost of hardware and installation because external connecting cables are no longer needed between the sensors and the data acquisition system. In addition, they allow for denser sensor networks and provide possibilities for distributed data acquisition and analysis (as opposed to the traditional approach where all the data are collected in one center). Denser arrays of sensors would improve the direct sensing capability of the monitoring system, while distributed data management would improve the reliability in data analysis. The current challenges are related to the power supply of wireless nodes and data losses during wireless transmission. However, ongoing research in power-harvesting technologies and signal transmission make wireless sensing a very promising tool for crack detection and characterization.

Micro-electromechanical systems can be equipped with integrated circuits and act as intelligent sensors, actuators, or both (e.g., [85,86]). More importantly, MEMS can be embedded into structures for subsurface crack detection. Although these lead to sophisticated devices with promising application potential, these types of possible sensors are somewhat limited. The most common example is accelerometers (e.g., [87]), to which modal analysis (having the limitations described previously) is applied.

Table IV. Indirect sensing methods (models and algorithms) for crack detection.

	Model-based approaches	Model-free approaches
Basic methodology	Measure changes of modal parameters (natural frequency, mode shape, damping, etc.) and compare with numerical models using data analysis methods	Statistical data analysis (pattern recognition) using sophisticated algorithms, without comparison with mathematical models
Examples of data analysis methods (the list is not exhaustive)	Model-based diagnostic approach and frequency domain identification; natural frequency and mode shape identification; static bending moment comparison; fatigue crack model with dimensional analysis, local stiffness reduction, and discrete spring model; 2D/3D complex model, uniform load surface technique combining generalized fractal dimension method and simplified gapped-smoothing method; local flexibility matrix model; boundary effect evaluation method utilizing operational deflection shape measured by scanning laser vibrometer; and so on.	Neural network method; Bayesian statistical system identification framework; Bayesian hierarchical model and artificial neural networks method; wavelets decomposition and wavelet coefficients analysis; Hilbert transform and Kalman filter; genetic algorithms; and so on.
Advantage	Limited number of sensors, contained cost, simple deployment, relatively simple data analysis	Limited number of sensors, contained cost, simple deployment, no need for numerical modeling
Challenge	Low sensitivity to minute crack, low accuracy in damage localization, on-site application (interference with environmental noise)	Low sensitivity to minute crack, damage localization practically impossible, on-site application (interference with environmental noise), complex data analysis

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Recently, strain sensors were also developed with MEMS (e.g. [86]). Although low-cost MEMS are emerging, the need for special processing steps implies increased costs.

Other advanced approaches such as piezoelectric paints (e.g., [88]), carbon nano-tube coatings (e.g., [89,90]), and embedded nano-sphere (e.g., [91]) are promising, but these are still not mature enough for applications to large-scale structures. Sensing skins based on carbon nanotubes are being developed for 2D strain-field monitoring [92]. Sensing skins utilize electrical impedance tomographical conductivity mapping techniques and provide a 2D mapping of damage [92]. Besides strain-field assessments, sensing skins can be made multi-functional (so that they can detect other defects such as corrosion) [93] and interfaced using wireless nodes [94]. Another promising approach is the use of multi-functional materials with self-sensing properties [95]. Recent discoveries point out that the electrical properties of cementitious materials can be used to detect and locate defects such as cracks [96,97]. Successful developments and large-scale implementations could practically transform an entire structure into a sensor.

A novel sensing technology based on LAE [98] can potentially be a good tool for crack detection and characterization for large-scale structures. Current direct sensing techniques that are based on strain measurements either monitor cracks at one point or segment (e.g., short-gage and long-gage sensors) or in one dimension (1D, e.g., distributed sensors). LAE sensing sheets can be treated as a quasi-distributed sensor in two dimensions. The concept of sensing sheet is shown in Figure 16.

Large-area electronics is an emerging technology that allows a broad range of electronic devices to be integrated on low-cost plastic sheets [99,100]. Through the use of micro-fabrication techniques, thin-film transducers (including pressure sensors, vapor sensors, particle sensors, etc.) have been demonstrated, which can be formed into dense arrays spanning large areas (i.e., tens of square meters). An important benefit of LAE is that it enables the integration of functional thin-film transistors (TFTs). This means that basic circuit functionality is available to facilitate readout from the large number of sensor channels. While these TFTs can provide basic functionality, the device-level characteristics that make them compatible with flexible, large-area substrates also severely limit their energy efficiency. Thus, large-scale processing over the sensor channels is not viable. Standard electronics technologies (e.g., based on integrated circuits) have achieved very high efficiency for instrumentation and processing because of nearly five decades of Moore's-law scaling. Researchers are now exploiting the basic possible functionality through the TFTs to create specialized interfaces between the large-area sensors and a potentially large number of readout and processing devices. The individual sensors are read by embedded integrated circuits with wireless and computational capabilities, taking advantage of wireless sensing and distributed data management. Finally, the protective layer is made of flexible photovoltaic, which provides the power needed for the functioning.

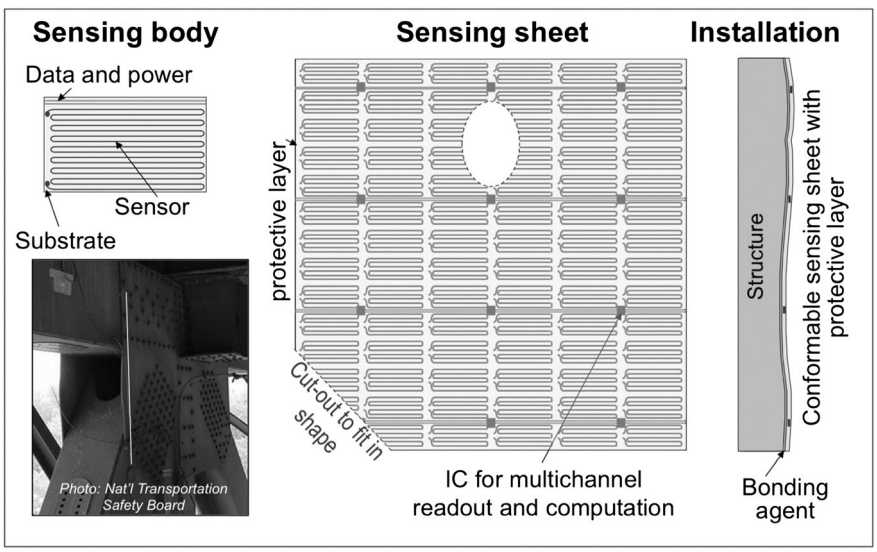

Figure 16. Concept of a sensing sheet.

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Figure 17 shows 2D strain measurements, illustrating how these provide a reliable modality (i.e., offering high sensitivity and specificity) for the detection and localization of cracks. In short, LAE not only can provide a low-cost crack monitoring through a dense and expansive array of sensors but also can overcome the robustness and reliability limitations affecting current SHM technologies.

In addition, noncontact sensing techniques have also become promising crack detection tools in recent years. Lecompte *et al*. [101] proposed a crack detection approach with two different camera techniques: a light-emitting diode combined with a charge-coupled device technique (LED-CCD) and the digital image correlation technique (DIC). Both techniques were used to detect cracks on the surface of a concrete beam subjected to flexural loading. They measured displacements in several discrete points on the surface and calculated the deformation using the Green–Lagrange strain expression. It was shown that these two techniques are complementary: LED-CCD is good for measuring a few data points in a long time, while DIC performs well for a dense file of data points in a short time. With both techniques, they confirmed the possibility of detecting the appearance and evolution of cracks even before cracks are visually observable. Jahanshahi and Masri [102], and Uhl *et al*. [103] also presented other vision-based techniques, such as optical instrumentation (e.g., digital cameras), and image processing and computer vision approaches for non-destructive evaluation of crack damage in real bridge structures.

Liu *et al*. [104] and Liu and Chen [105] introduced the Terrestrial 3D Light Detection and Ranging scanner (LiDar) as a remote sensing technique for health monitoring of existing and newly constructed bridges. They proposed an automated bridge evaluation algorithm called LiDar bridge evaluation (LiBE) for on-site bridge monitoring, which measured the damage area and operated robustly under changing environmental conditions. This technique could be applied in crack detection, bridge clearance, and static defection measurement and is especially useful when accurate measurement of bridge geometry is impossible. However, resolution and complex data analysis are main challenges of this

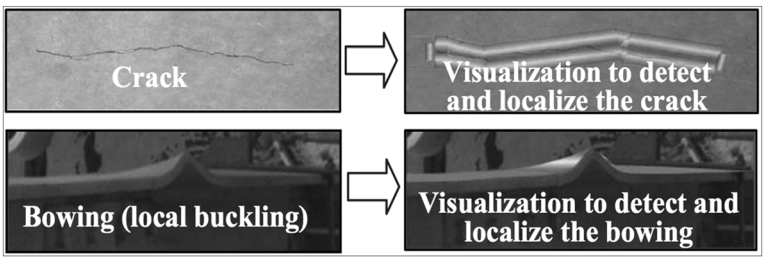

Figure 17. Illustration of the potential for direct damage detection using high-resolution 2D strain-field measurements. Left-side images show the damage, and right-side images show the expected measurement.

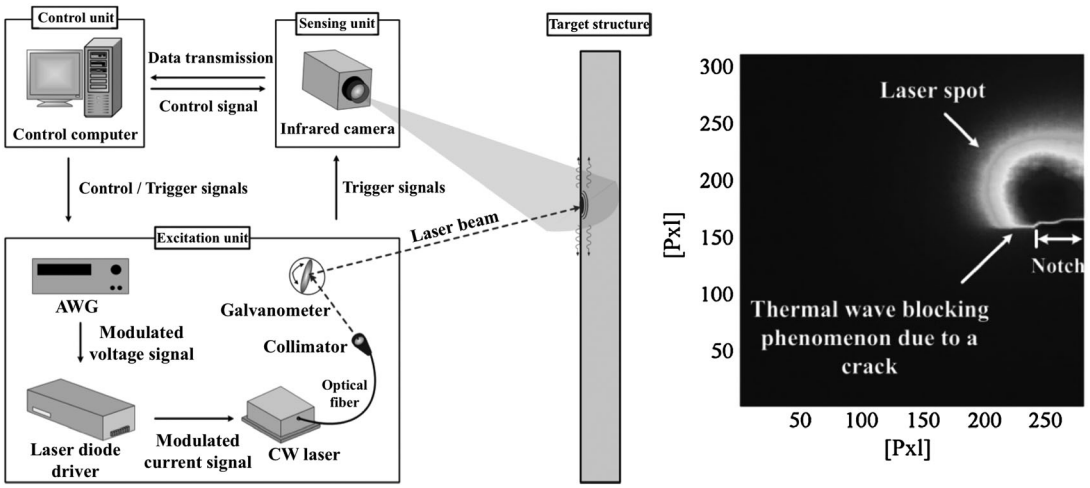

Figure 18. Noncontact laser lock-in thermography (LLT) technique; left: LLT system; right: LLT image for crack detection in a plate structure [108].

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Table V. Advantages and challenges of the emerging sensing technologies for crack detection and characterization.

Emerging technologies	Wireless networks	MEMS	Sensing skin	Self-sensing materials	LAE sensing sheet	Noncontact sensing techniques
Advantage	Cost reduction of hardware and installation (no connecting cables), distributed data acquisition and processing	Availability for subsurface crack detection	High spatial resolution (2D strain-field monitoring), corrosion detection, wireless	High spatial resolution (2D strain-field monitoring)	Low cost, high spatial resolution (2D strain-field monitoring), multi-functional, wireless	High spatial resolution and sensitivity, accurate crack characterization
Challenge	Power supply and data transmission losses	Sensor packaging, on-site application (mostly used as accelerometer) and high cost	Immature for widespread applications	Immature for widespread applications	Immature for widespread applications	Real-time crack detection (mostly used as NDT and not SHM), complex data analysis
Direct or indirect sensing	Enables direct or indirect sensing depending on type of sensor attached to wireless nodes	Enables direct or indirect sensing depending on type of sensor	Direct	Direct	Direct	Direct

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technology. Several field applications were discussed: Imperial Highway bridge in Los Angeles, steel bridge over I-77 in Mecklenbury Country, and concrete bridge over US-74. The capability of accurate damage detection from the LiBE system was validated through the aforementioned bridge applications.

Sohn [106] provided an overview of ongoing laser ultrasonic-based SHM researches for damage detection in civil, mechanical, and aerospace structures. Ultrasonic-based SHM techniques can detect very small defects even at a remote location on structure. Noncontact laser ultrasonic techniques, in particular, have better performance than the conventional ultrasonic techniques in three aspects: (1) they have higher spatial resolution; (2) they are baseline free and have less false positive detections; and (3) they are faster to deploy. These techniques have been extensively studied in the NDE domain, and Sohn [106] introduced them in SHM of various structures, such as aircraft, wind turbine, high-speed train, nuclear power plant, and a real bridge in South Korea. An *et al*. [107] introduced a laser ultrasonic-based crack visualization approach for an aluminum plate structure in laboratory, with a novel image processing technique. The main challenges of noncontact laser ultrasonic techniques come from the following facts: (1) it requires long time for scanning, more expensive hardware, and complex data analysis; (2) special surface treatment is needed; and (3) because of the strong laser, eye protection during testing should be provided.

Another laser technique appearing recently is based on laser thermography. An *et al*. [108] presented a noncontact laser lock-in thermography (LLT) technique for fatigue crack detection in a steel plate structure. The LLT technique dominates conventional laser thermography technique by the following facts: (1) the laser heat source can be located precisely even at a long distance away; (2) it is capable to inspect large structure; (3) no special surface treatment is required; and (4) environmental noises can be eliminated. With this technique, the authors proposed a holder exponent filter approach for crack identification, localization, and quantification. The fatigue crack length in the plate identified by LLT technique was close to the experimental results, which validated such a method for damage detection. Besides, this technique has the potential to be applied in real structures, such as bridges, buildings, train vehicles, and so on. An example of the LLT system and its application for crack detection is shown in Figure 18.

Advantages and challenges of the emerging sensing technologies are summarized in Table V. The potential of these emerging technologies for either direct or indirect sensing is also presented in Table V.

This paper focuses on sensing techniques for crack detection and characterization. However, it is also important in engineering practice to carry out long-term monitoring of the evolution of existing and already identified cracks. On the basis of literature review, and the authors' knowledge and experience, several of the techniques presented in this paper, such as vibrating-wire-based sensors, fiber-optic sensor (short-gage, long-gage, and distributed), coaxial cable, and wave-based techniques, have the capability and have already been applied in real projects for this purpose. All emerging sensing technologies also have the potential for long-term crack evolution monitoring.

6. CONCLUSIONS

Cracks are critical flaws that affect the behavior and durability of civil structures and infrastructure. Thus, reliable crack detection and accurate crack characterization are important objectives of SHM. Numerous crack types and causes are summarized in this paper. Various crack detection and characterization techniques are presented, including direct sensing approaches, indirect sensing approaches, and emerging technologies.

Two major groups of current approaches are identified: direct sensing and indirect sensing. Both have advantages and challenges. The main advantage of the direct sensing is its reliable crack detection and, to a certain extent, its accurate crack characterization. This statement is supported with results reported from real on-site applications, especially with strain sensing. However, the challenge of this approach is the need for dense arrays of sensors, which requires high hardware, installation, and data management costs. On the other hand, the main advantage of indirect sensing is its lower cost for hardware and installation. However, the challenge is its reliability in crack detection and accuracy in crack characterization when applied in real-life settings. The new and emerging sensing technologies, as well

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as never-ending developments in computational power, are all paving ways for overcoming the aforementioned challenges. The authors believe that the improved reliability, robustness, and accuracy in crack detection and characterization, combined with the low cost that these novel technologies can offer, have the potential to transform SHM and open the doors for its widespread application, vastly improving safety and reducing maintenance and life-cycle costs for civil infrastructure.

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